Status	Finished
Started	Sunday, 25 October 2026, 10:21 PM
Completed	Sunday, 25 October 2026, 10:30 PM
Duration	8 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O👎
- d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. Number values should only be present

  Number values should only be present
- b. There should be direct access to the middle element in any sublist
- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. The list must be sorted

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the right sublist
- b. In the left sublist ✓
- c. At the beginning of the list
- d. In the entire list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search can find multiple instances of the target element
d. Binary search does not require dividing the list

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. n ✓
d. log n

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search requires the list to be sorted.
c. Linear search is more efficient than binary search on large lists.
d. Linear search divides the list into halves.

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(log n)
c. O(n log n)
d. O(1)

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. When the item is not the array at all

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n

b. n-1

c. n/2

d. n ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements

b. When the list is sorted ✓

c. When the list is very small

d. When the list is unsorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎

b. O(1) ✓

c. O(n log n)

d. O(log n)

Question **12**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search ends successfully ✓
- b. The list is sorted
- c. The search continues in the left sublist
- d. The search continues in the right sublist

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. An error is raised
- c. The last element is returned
- d. The first element is returned

Question **14**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the search algorithm finishes
- d. When the element is found in the list ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Hash search
- b. Linear search ✓
- c. Both (1) & (3)
- d. Binary search